

data['Ticker'].astype('category') (.cat.as_ordered()) | import matplotlib.pyplot as plt

df.dtypes

ss.unique()

data.groupby('Ticker')['Date'].count() | plt.boxplot(...) plt.hist(...)

value_counts().sort_index() | data['Date'] = pd.to_datetime(data['Date'])

np.where(data.isnull())

sns.regplot

sns.heatmap

sns.scatterplot

sns.countplot

sns.boxplot

from statsmodel.stats import outliers_influence

levs = outliers_influence.OLSInfluence(model).hat_matrix_diag

formal = '%Y %m %d'

df.plot(kind='bar', stacked=True)

accuracy: $E(\hat{\theta}) = \theta$

unbiased: $E(\hat{\theta}) = \theta$

precision: $V(\hat{\theta})$

SE: $\sqrt{V(\hat{\theta})}$

press = ((model.resid / (1 - levs)) ** 2).sum()

$1/SE = E((\hat{\theta} - \theta)^2)$

$NVUE = V(\hat{\theta}) + bias^2(\hat{\theta})$

$bias = E(\hat{\theta}) - \theta$

Case 1 $\hat{p} = X/n$, $SE \approx \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

Rule of thumb: $np \geq 5$, $n(1-p) \geq 5$

Case 2 uncorrelate $X_1, X_2, \dots, X_n$, $E(X_i) = \mu$, $V(X_i) = \sigma^2$

$$s^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{(n-1)}$$

Case 3 $X_1, X_2, \dots, X_n$ iid $N(\mu, \sigma^2)$, $\Rightarrow \bar{X} \sim N(\mu, \sigma^2/n)$

① $\bar{X} \perp s^2$ ② $\frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$

③ $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim Z$ ④ $\frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t(n-1)$

$\hat{\sigma}^2 = RSS/(n-p-1)$ (simple linear regression $p=1$)

$\frac{\hat{\beta}_i - \beta_i}{SE(\hat{\beta}_i)} \sim t(n-p-1)$

CI: $\hat{\beta}_i \pm t_{\alpha/2, n-p-1} SE(\hat{\beta}_i)$

AIC = $n \log(RSS) + 2p$ minimize

sm.OLS(endog=y, exog=X)

$PRESS = \sum_{i=1}^n (y_i - \hat{y}_{(i)})^2 = \sum \frac{\hat{\epsilon}_i^2}{1 - h_{ii}} \geq RSS$

$X_i \sim \text{Exp}(\beta) \Rightarrow \sum_{i=1}^n X_i \sim \Gamma(n, \beta)$

$\chi^2(k) \sim \Gamma(\frac{k}{2}, \frac{1}{2}) \sim \sum_{i=1}^k Z_i^2 = X$

$E(X) = k$, $V(X) = 2k$

$\chi^2(v)$ $T = \frac{Z}{\sqrt{X/v}}$ $E(T) = 0$ $V \geq 2$ $V(T) = \frac{v}{v-2}$ $v \geq 3$

Case 3 $n < 30$ $100(1-\alpha)\%$ CI for μ : $\bar{X} \pm t_{\alpha/2, n-1} (s/\sqrt{n})$ from scipy.stats
 Case 2 $n > 30$ $\bar{X} \pm z_{\alpha/2} (s/\sqrt{n})$ import t, norm

Case 1 $np \geq 5, n(1-p) \geq 5$

from statsmodels.graphics.gofplots import qqplot
 from scipy.stats.distributions import import t, norm

$x = t.rvs(30, size=100)$
 $qqplot(x, norm, line='45', fit=True)$

MOM: solve $M_k = \sum_{i=1}^n x_i^k / n = E(X^k)$



$p: \hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

$\text{Exp}(\lambda) \hat{\lambda}_{MLE} = \frac{1}{\bar{X}} = \hat{\lambda}_{MOM}, SE = \frac{1}{n\bar{X}^2}$

$U(0, \theta) \hat{\theta}_{MLE} = X_{(n)} \quad \hat{\theta}_{MOM} = 2\bar{X}$

M/σ
 signal to noise ratio

Invariance of MLE: $g(\hat{\theta}_{MLE}) = \text{MLE for } g(\theta)$

$I(\theta_0) = E\left(-\frac{\partial^2 \log f(X; \theta)}{\partial \theta^2}\right)$ (efficiency)
 Cramer-Rao

Asym. Normality: $\sqrt{n}(\hat{\theta} - \theta_0) \rightarrow N(0, I^{-1}(\theta_0))$ ($\hat{\theta} \sim N(\theta_0, I^{-1}(\hat{\theta})/n$) lower bound

Delta method $Y_n \xrightarrow{D} N(\mu, \sigma_n^2)$ $\sigma_n^2 \downarrow 0 \Rightarrow g(Y_n) \sim N(g(\mu), (g'(\mu))^2 \sigma_n^2)$ $\sigma_n^2 = \sigma^2/n$ when $Y_n = \bar{X}_n$

Fisher Info matrix:

$g(\hat{\theta}) \sim N(g(\hat{\theta}), (g'(\hat{\theta}))^2 (1/n I(\hat{\theta})))$ Delta + MLE

$I_{ij}(\vec{\theta}) = E\left(-\frac{\partial^2}{\partial \theta_i \partial \theta_j} \log f(X; \vec{\theta})\right)$

$\begin{pmatrix} g_1(Y_n) \\ g_2(Y_n) \\ \vdots \\ g_m(Y_n) \end{pmatrix}$

$m \leq p \quad G_{ij} = \frac{\partial g_i(Y)}{\partial y_j}, \quad G \text{ full rank}$

Delta method m-Dim

$Y_n \rightarrow N_p(\mu, \Sigma/n), \quad g =$

$\Rightarrow g(Y_n) \sim N_m(g(\mu), G \Sigma G^T/n)$

sample cov $S_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$ $\hat{\beta}_1 = r_{xy} \left(\frac{s_y}{s_x}\right)$: unbiased, $SE = \frac{\hat{\sigma}}{s_x \sqrt{n-1}}$, $T = \frac{\hat{\beta}_1 - \beta_{1,true}}{SE} \sim t(n-2)$

$S_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

sample corr $r_{xy} = \frac{S_{xy}}{s_x s_y}$

$\hat{\sigma}_{MLE}^2 = RSS/n$

$RSS = \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$ unbiased: $\hat{\sigma}^2 = RSS/(n-2)$

import statsmodels.api as sm
 model = sm.OLS.from_formula('y ~ x1 + x2 + x3', data=df)
 model.params.summary()
 bse
 resid
 fitted values
 $V(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$
 $X^T \hat{\epsilon} = 0$
 $\hat{Y} = X \hat{\beta} = H Y$
 $\hat{\epsilon} = (I - H) Y = Y - \hat{Y}$
 $V(\hat{\epsilon}) = \sigma^2 (I - H)$